B.Sc. (Data Science)

DS101T : Problem Solving and Python Programming

Unit 2: Python Libraries

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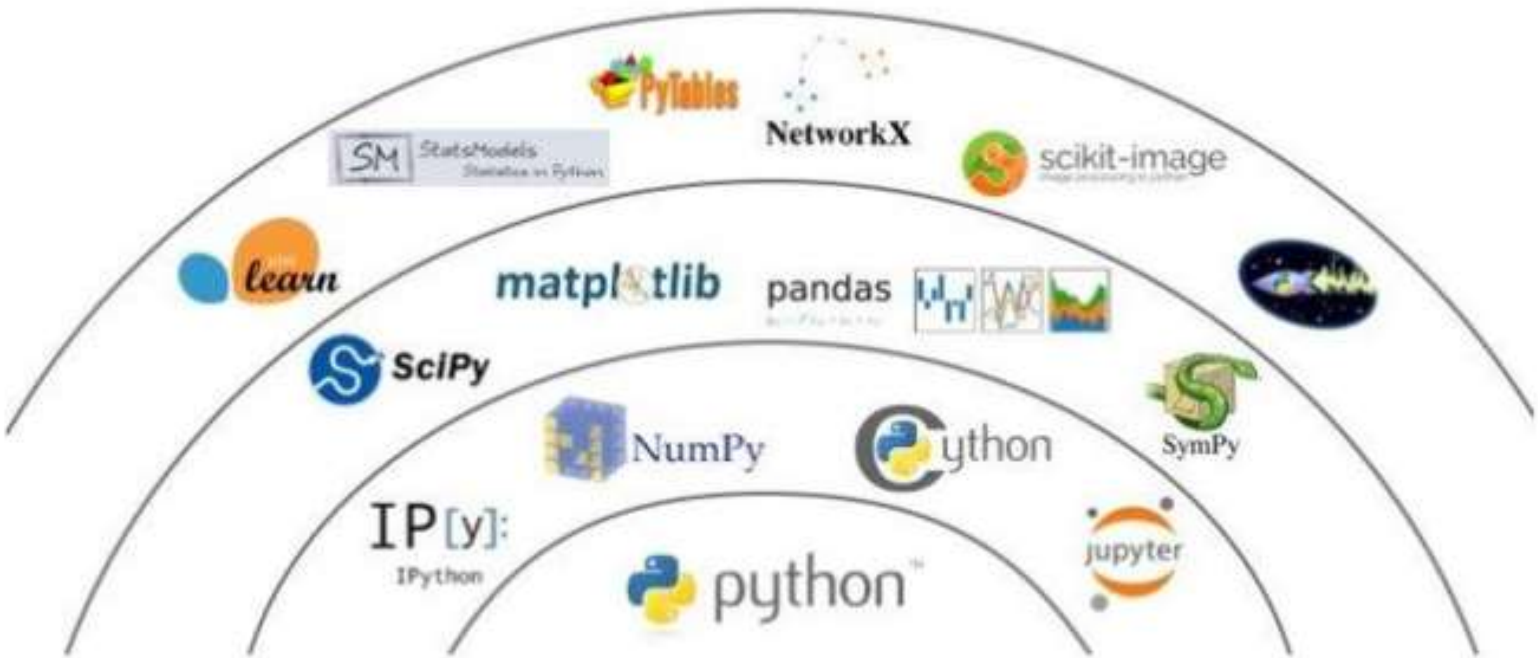
Unit 2: Python Libraries

- 2.1 Introduction to Python Libraries
- 2.1.1 Statistical Analysis- NumPy, SciPy, Pandas, StatsModels
- 2.1.2 Data Visualization- Matplotlib, Seaborn, Plotly
- 2.1.3 Data Modelling and Machine Learning- Scikit-learn, XGBoost, Eli5
- 2.1.4 Deep Learning- TensorFlow, Pytorch, Keras
- 2.1.5 Natural Language Processing (NLP)- NLTK, SpaCy, Gensim

Python Libraries for Data Science

- There are several libraries that have been developed for data science and machine learning.



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Python Libraries for Data Processing

- NumPy
- SciPy
- Pandas
- Statsmodels

NumPy, SciPy

- **NumPy** stands for Numerical Python.
  - This library is mainly used for working with arrays.
  - Operations include adding, slicing, multiplying, flattening, reshaping, and indexing the arrays.
  - This library also contains basic linear algebra functions, Fourier transforms, advanced random number capabilities etc.

- **SciPy** stands for Scientific Python.
  - SciPy is built on NumPy.
  - It is one of the most useful library for variety of high level science and engineering modules like discrete Fourier transform, Linear Algebra, Optimization and Sparse matrices.

Statsmodels

- **Statsmodels** for **statistical modeling**.
  - Statsmodels is part of the Python scientific stack, oriented towards data science, data analysis and statistics.
  - It allows users to explore data, estimate statistical models, and perform statistical tests.
  - An extensive list of descriptive statistics, statistical tests, plotting functions, and result statistics are available for different types of data and each estimator.

Pandas

- **Pandas** is used for structured data operations and manipulations. Pandas provides various high-performance and easy-to-use data structures and operations for manipulating data in the form ofnumerical tables and time series. It provides easy data manipulation, data aggregation, reading, and writing the data as well as data visualization. Pandas can also take in data from different types of files such as CSV, excel etc.or a SQL database and create a Python object known as a data frame.
- The two main data structures that Pandas supports are:

  Series: Series are one-dimensional data structures that are a collection of any data type.

  DataFrames: DataFrames are two-dimensional data structures which resemble a database table or say an Excel spreadsheet.

- **NumPy:**
  Used for preprocessing data and creating input arrays for machine learning models.

- **Pandas:**
  Handles tabular data, performs aggregations, and cleans datasets.

- **SciPy:**
  Solves advanced problems like integration, interpolation, and differential equations.

Parameter	**NumPy**	**Pandas**	**SciPy**
Focus	Numerical and array computations.	Data manipulation and analysis.	Advanced scientific computations.
Core Structure	ndarray for multi-dimensional data.	DataFrame and Series for tabular data.	Modules for optimization, integration, and more.
Ease of Use	Best for numerical operations.	Ideal for structured data.	Requires domain knowledge for usage
Integration	Integrates with Pandas, Matplotlib, etc.	Built on NumPy.	Works with NumPy arrays.
Applications	Matrix operations, broadcasting.	Data analysis, cleaning, visualization.	Optimization, signal processing, etc.